

SERVICE MANUAL



Document 2381683-9

Revision 02

**GE Signa[®] Excite[™] 3.0T
8-CHANNEL TORSO ARRAY**

GE Catalog Part Number: M1385AY

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Language Policy For Service Documentation (Dir. 2128126)

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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 8-Channel Torso Array, refer to the coil labels (see *Figure 1*).

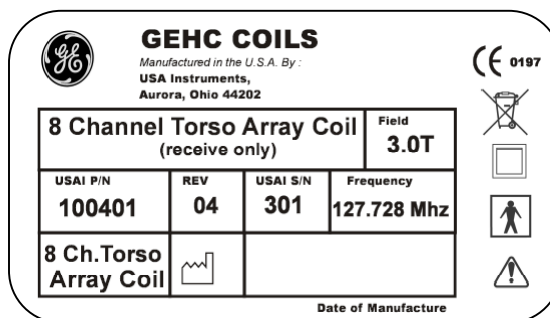


Figure 1: Coil label.

Figure 2 shows a picture of the 8-Channel Torso Array.



Figure 2: The 8-Channel Torso Array.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	N/A	U1-100401	1
1	Operator's Guide Packet	2422146-500	N/A	1
1	Service Manual	2381683-9	N/A	1
2	Patient Liner	N/A	U1-502548	1
2	Table Pad	N/A	U1-150485	1
2	Spacer Pad	N/A	U1-150486	1
2	Pad, Lower	2405545	N/A	1
3	Phantom	2381683-3	U1-150513	1
3	Phantom Positioner	N/A	U1-150515	1

1-2 Compatibility

This coil is compatible with the Signa[®] Excite[™] 3.0T MR system.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422146.

Signa[®] Excite[™] 3.0T Service Methods CD, 2389160-1.

1-4 Environmental Requirements

Storage Requirements

The coil should be stored in the scanner room.

Dimensions

43.3" W x 15.0" D x 13.0" H (110.0 cm x 38.1 cm x 101.9 cm)

Weight

14.0 lbs (6.4 kg)

1-5 Theory of Operation

The block diagram of the 8-Channel Torso Array is shown in *Figure 3*. *Figure 4* shows the block diagram of the circuit layout. Also refer to the schematic in the Appendix. The receive-only coil is designed to give optimum signal to noise ratio and uniform coverage of the abdominal and pelvic regions, including the liver, spleen and renal arteries. The coil package consists of anterior and posterior sections, which are positioned above and below the region of interest.

The coil is actively decoupled from the RF transmit coil during transmit by means of multiple RF choking circuits or RF chokes. Each coil element has three active decoupling circuit elements and two passive decoupling circuit elements, which can be switched on actively or passively. The passive RF chokes are switched on using small-signal diodes. The small-signal diodes are turned on by the induced RF voltage in the chokes, coupled from the transmit field (passive coupling). Each small-signal diode is turned on when the induced RF voltage reaches about 0.5 volt. The active RF chokes are switched on using PIN diodes. The PIN diodes are turned on by a forward biasing DC current source supplied by the MRI system (active decoupling). When either the PIN diodes or the small-signal diodes are turned on, the RF choking circuits become very high impedance blocks (typically above 3 k-ohm), compared to the other circuit elements (less than 70 ohm). These high impedance elements segregate the coil circuitry into several isolated electrical segments, preventing any current flow in the coil circuit that might be induced by the body coil transmission. In the event that one of the diodes fails, the signal diode will form a short circuit and permanently de-resonate the coil.

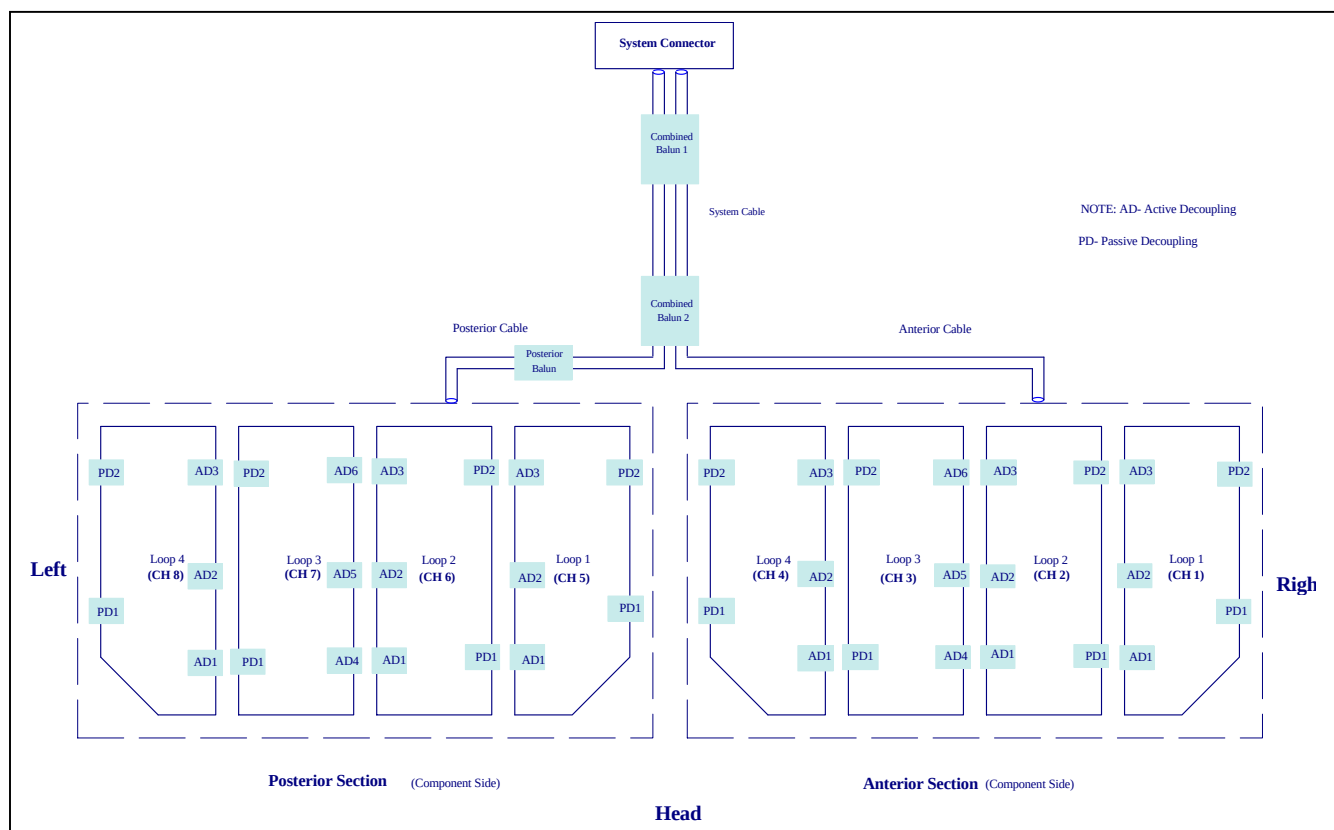


Figure 3: Block diagram of the coil.

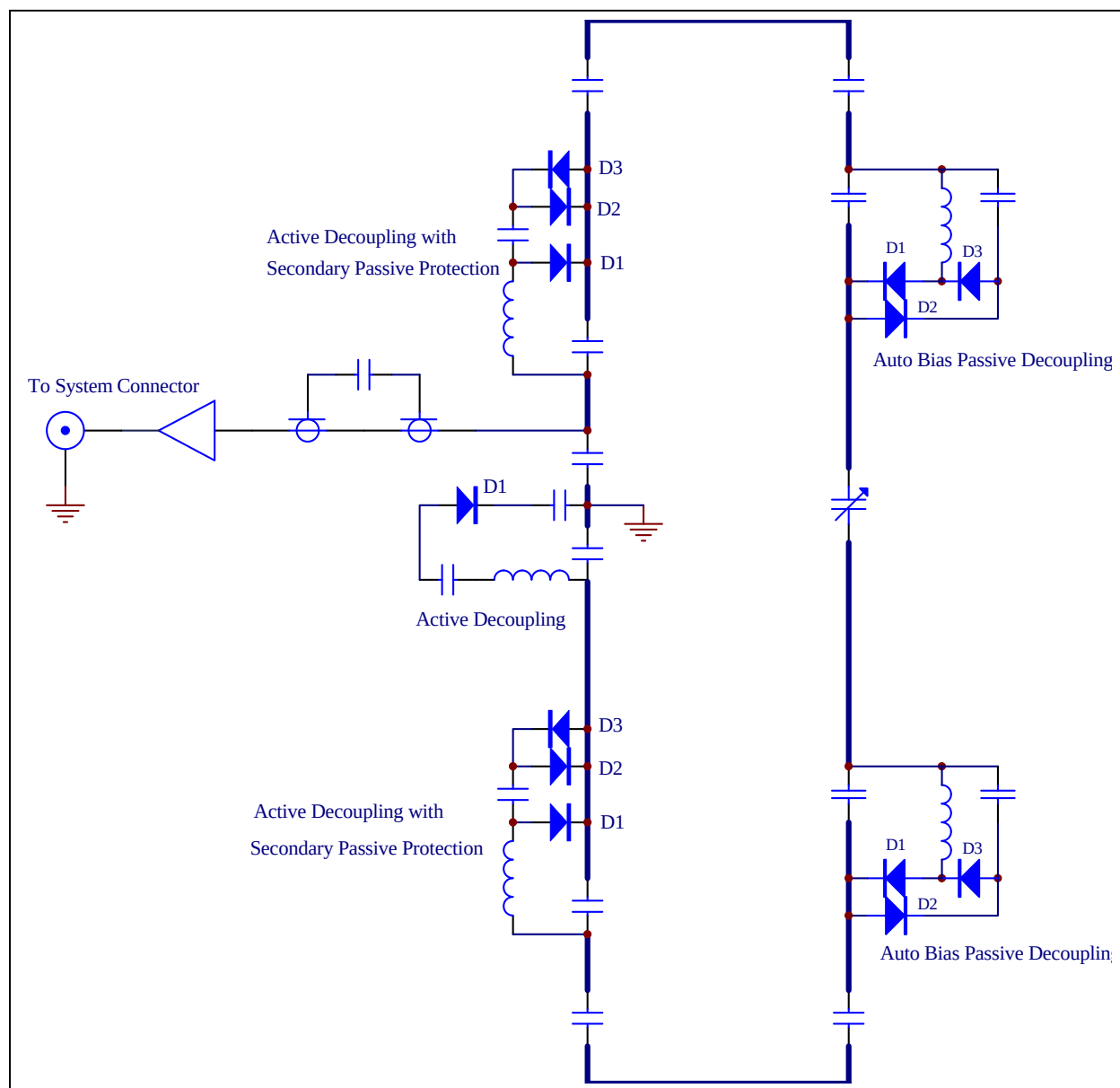


Figure 4: Block diagram of the circuit layout.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

The name for this coil is 8US TORSOPA. Add the coil using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager. For reference, the configuration information is included in the Appendix; however, that configuration is only current as of the printing of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; RF Path; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-1

Description	GE Part #	Qty
Phantom	2381683-3	1
Phantom Positioner	2381683-4	1

3-2-2 Data Collection

Step 1 – Place the 8-Channel Torso Array on the patient cradle (see *Figure 5*). Note the serial number of the coil.

Step 2 – Position the phantom positioner on the coil as shown in *Figure 6*.



Figure 5: 8 Channel Torso coil setup.



Figure 6: Phantom Positioner setup.

Step 3 – Position the phantom on the phantom positioner as shown in *Figure 7*.

Step 4 – Place the anterior section on the phantom and fasten the Velcro straps on both sides (see *Figure 8*).

Step 5 – Establish an axial landmark on the line on top of the coil. Advance to scan.

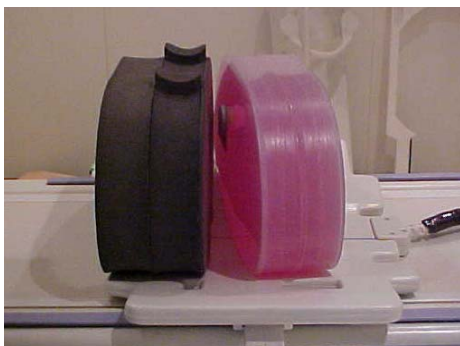


Figure 7: Phantom setup.



Figure 8: Anterior coil setup.

Step 6 – Move back to the operator workspace and end the previous exam (if required).

Step 7 – For 12.x or earlier systems: From the Common Service Desktop, select **Image Quality**, and then **Coil SNR**. Then select **Click here to start this tool**.

For 14.x or later systems: From the Common Service Desktop, select **Image Quality**, and then **MCQA Tool**. Then select **Click here to start this tool**. For additional information regarding MCQA, refer to the 14.x or later Service CD at the following location: troubleshooting/Image Quality/Multicoil Quality Assurance.

Step 8 – Select the **[8-Channel Torso Coil]** from the Coil Name selection menu on the Multicoil SNR Tool Control window (see *Figure 9*).

Step 9 – Enter the Coil serial number.

Step 10 – Enter the number of repetitions in the Passes field. Typically 1 pass is sufficient.

Step 11 – Hit the “**Start Scan**” button. The system will acquire the images and automatically start data analysis.

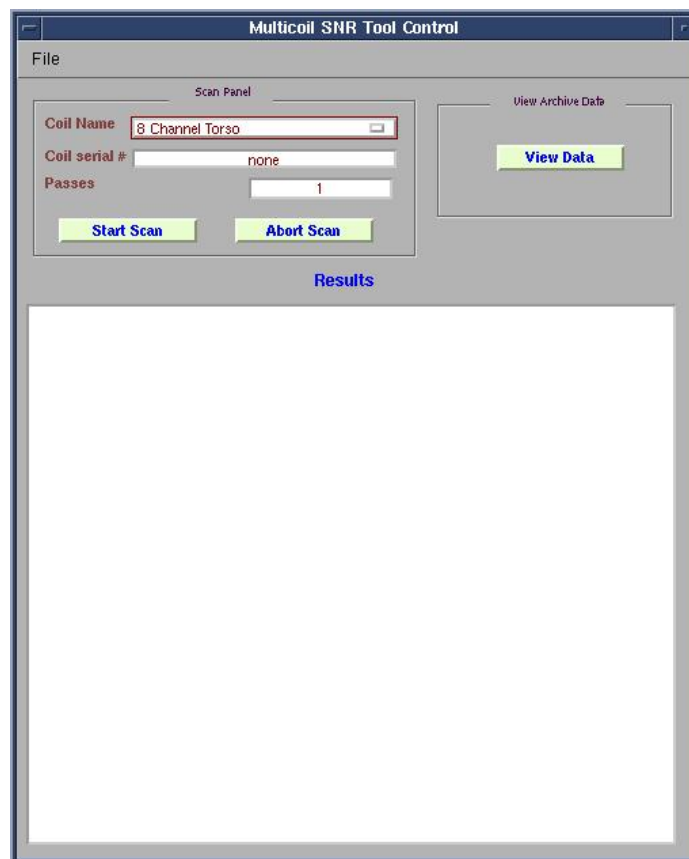


Figure 9: Multicoil SNR Tool Control window.

3-2-3 Data Analysis

When the scan completes, the data will be displayed in the Multicoil SNR Tool Control window (see *Figure 10*). Each channel along with the composite image is assigned a pass/fail value.

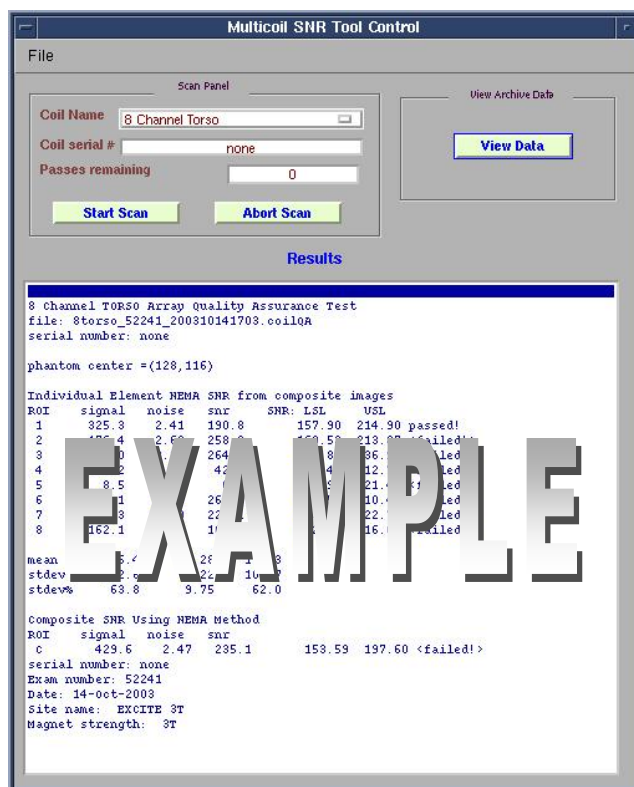


Figure 10: Results displayed in the Multicoil SNR Tool Control window.

COMPOSITE SNR ACCEPTANCE LIMITS – TABLE 3-2-3-A

Parameter	Mean	USL	LSL
Composite SNR	211.4	321.99	98.01

INDIVIDUAL ELEMENT SNR LIMITS – TABLE 3-2-3-B

Parameter	Mean	USL	LSL
Ch 1 Individual Element Composite	369.4	397.50	202.50
Ch 2 Individual Element Composite	353.2	445.17	214.83
Ch 3 Individual Element Composite	496.4	581.83	368.17
Ch 4 Individual Element Composite	506.9	560.19	339.81
Ch 5 Individual Element Composite	512.7	569.22	330.78
Ch 6 Individual Element Composite	424.2	572.96	367.04
Ch 7 Individual Element Composite	448.3	579.46	340.54
Ch 8 Individual Element Composite	578.2	626.82	393.18

Note: The SNR specification limits are current at the release of this manual. The most current SNR specification limits are included in the automated SNR measurement tool.

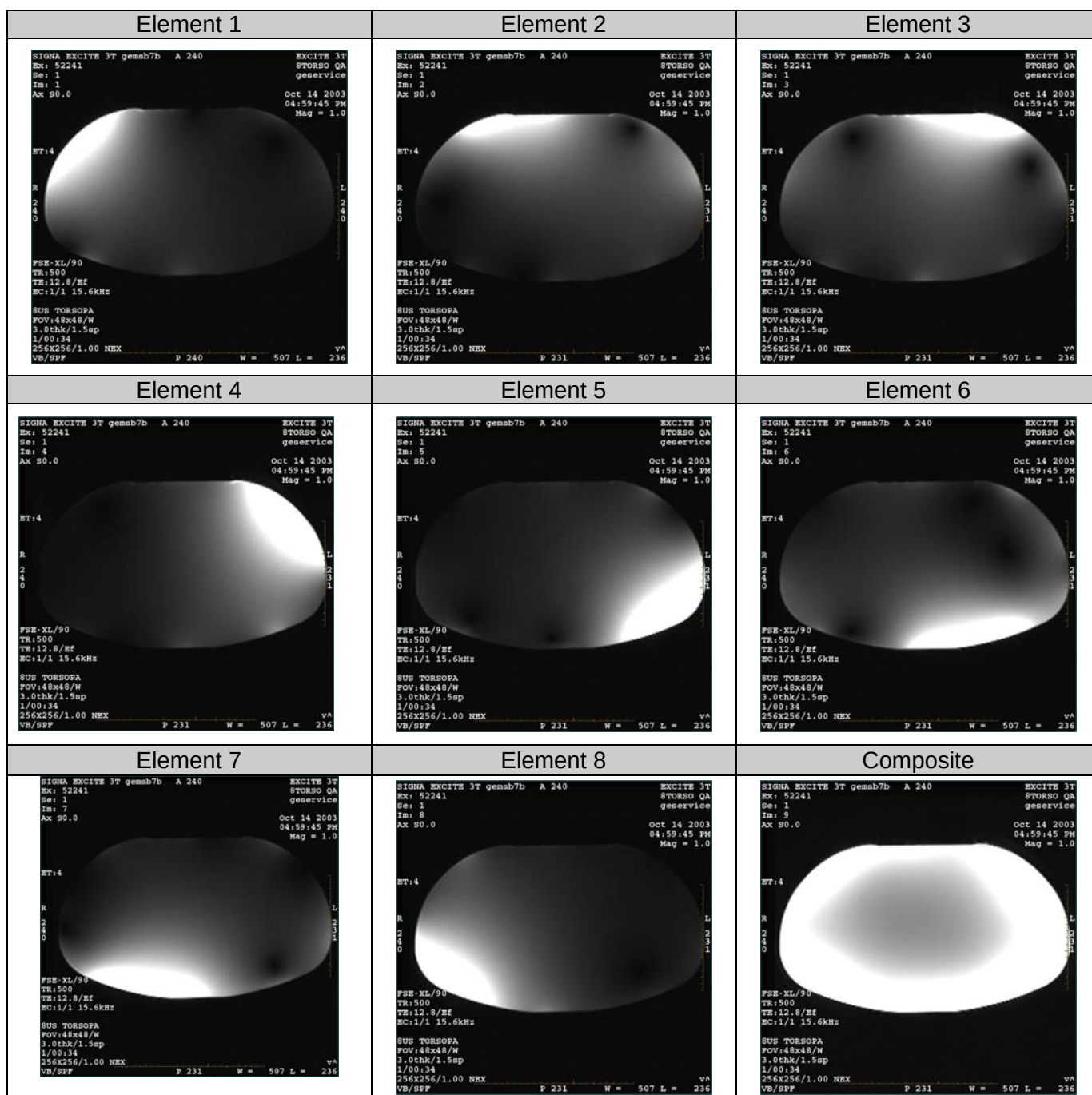


Figure 11: Individual elements.

To View Archived Data Files

Step 1 – Click on the “**View Data**” button in the Multicoil SNR Tool Control window (see Figure 9).

Step 2 – Select the [FileTab] in the now open Multicoil SNR Display Control window (see Figure 12).

Step 3 – Turn Expert Mode on.

Step 4 – Select the 8-Channel Torso coil in the coil selection menu.

Step 5 – Click the “**Plot Data**” button.

Step 6 – The Data Plot Control window opens (see *Figure 13*).

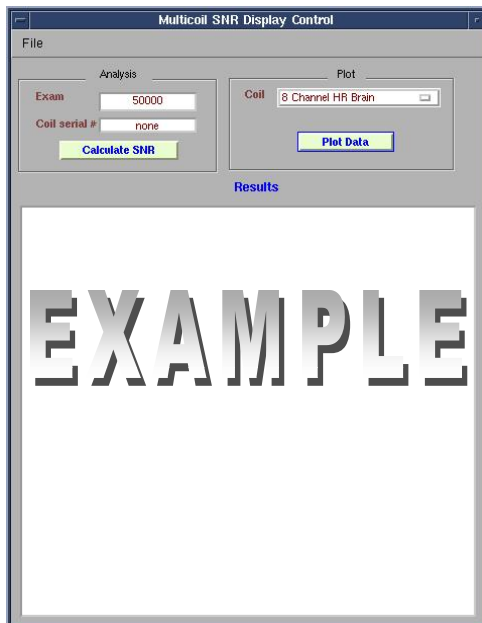


Figure 12: Multicoil SNR Display Control window.

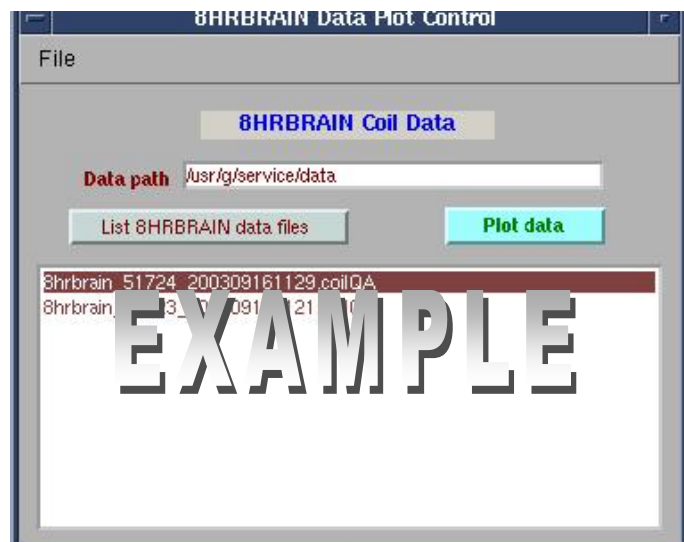


Figure 13: Data Plot Control window.

Step 7 – Select the desired file and click “**Plot Data**”.

Step 8 – A plot window with SNR, Signal and Noise plots will open (see *Figure 14*).

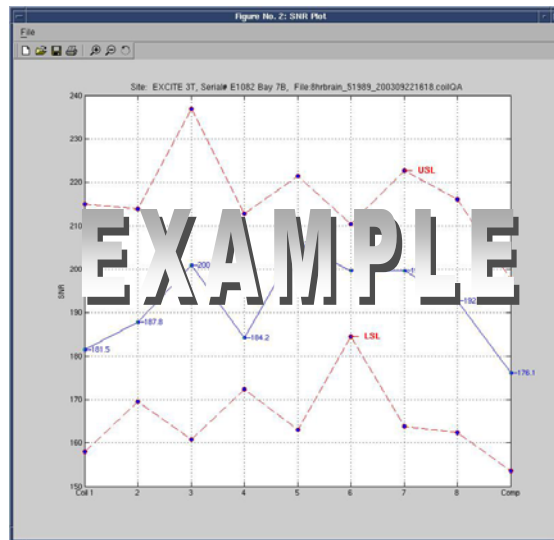


Figure 14: Nominal Individual Element results.

3-2-4 Test Explanation and Results Interpretation

The 8-Channel Torso Array is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses. Separate signal and noise images are evaluated for individual element signal, noise and SNR. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference.

3-3 External Cable Check

If the signal on any channel is low, refer to *Figure 15* to ensure that none of the B, C or D row pins are bent.

3-4 PIN Diodes Check

No diode checks can be made from the connector. All PIN diode checks must be made individually with the coil open.

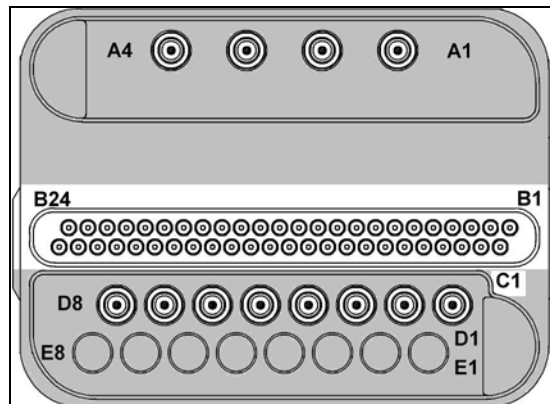


Figure 15: 48-pin Hypertronics connector.

3-5 Mechanical Hardware Check

None.

3-6 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	Replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs	Correct MRI system problem.
Coil ID issues.	From CSD, select Diagnostics, Manual Tests, SRI Functional Checks, and run the SRI Coil ID test. Compare the coil ID value found to that listed in Section 7-2 of this manual.	If the value does not match, or no coil ID is found, replace the coil.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Check QA results from Data Sheet	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise std. dev. measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual. Check pins for preamp bias.	Correct MRI system DC bias problem – otherwise replace the coil.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.

CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: a ten percent bleach solution (some discoloration may occur) or warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #	USAI Part #
Coil w/cable	2381683-2	U1-165149
Phantom	2381683-3	U1-155190
Phantom Positioner	2381683-4	U1-155191

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #	USAI Part #
Patient Liner	2381683-5	U1-155192
Table Pad	N/A	U1-155193
Spacer Pad	N/A	U1-155194
Lower Pad	2406572	N/A

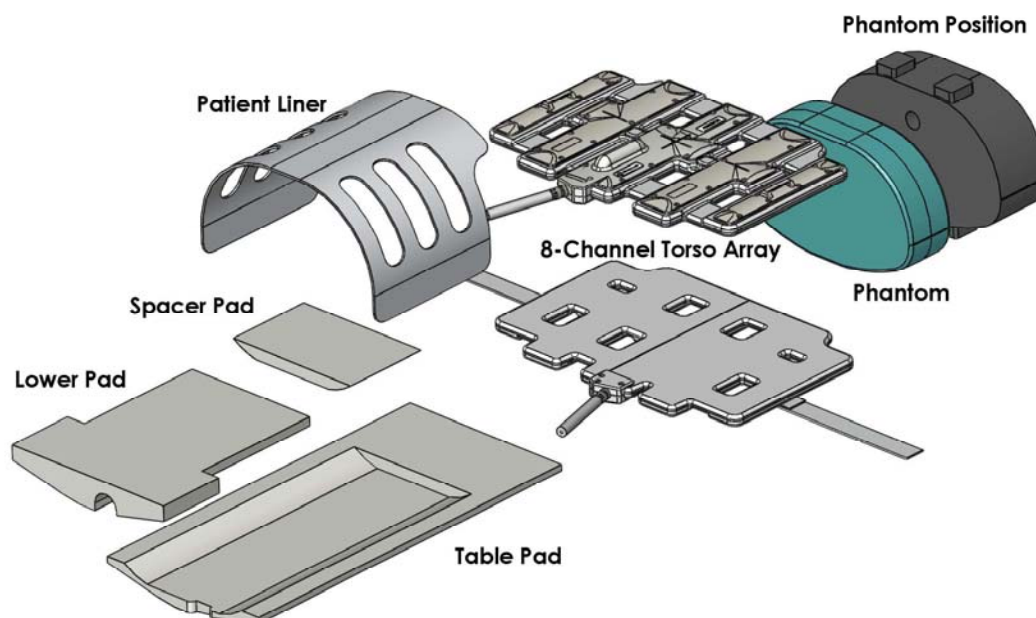
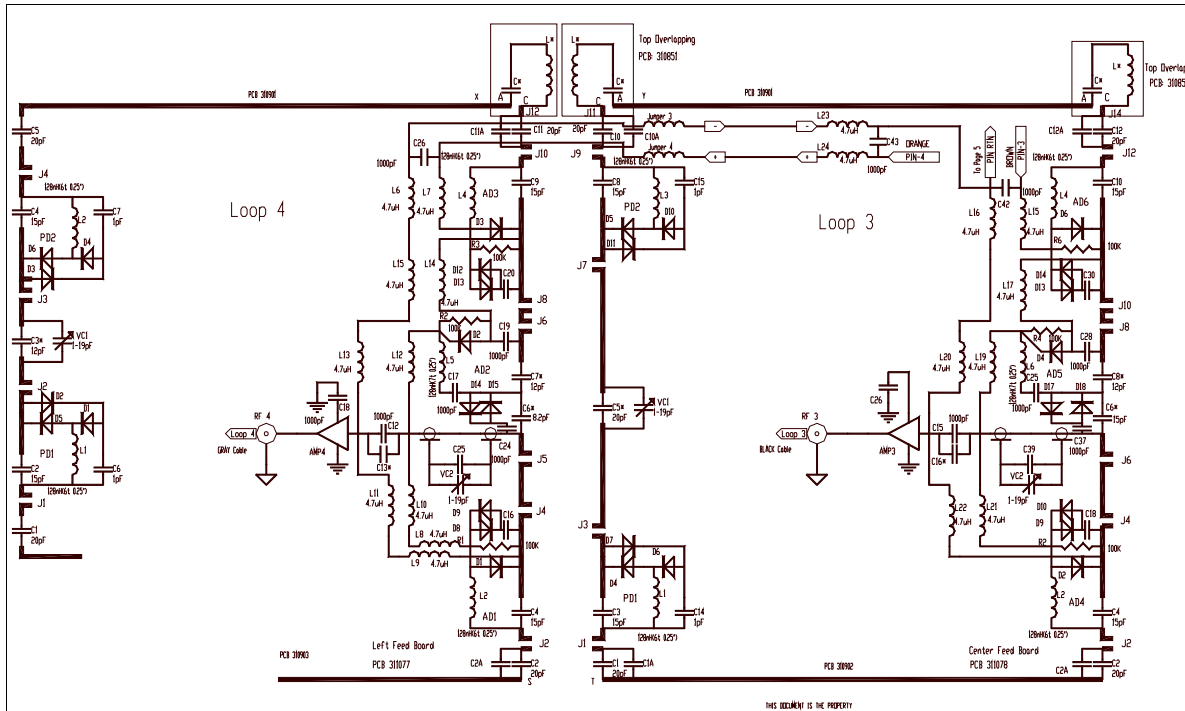


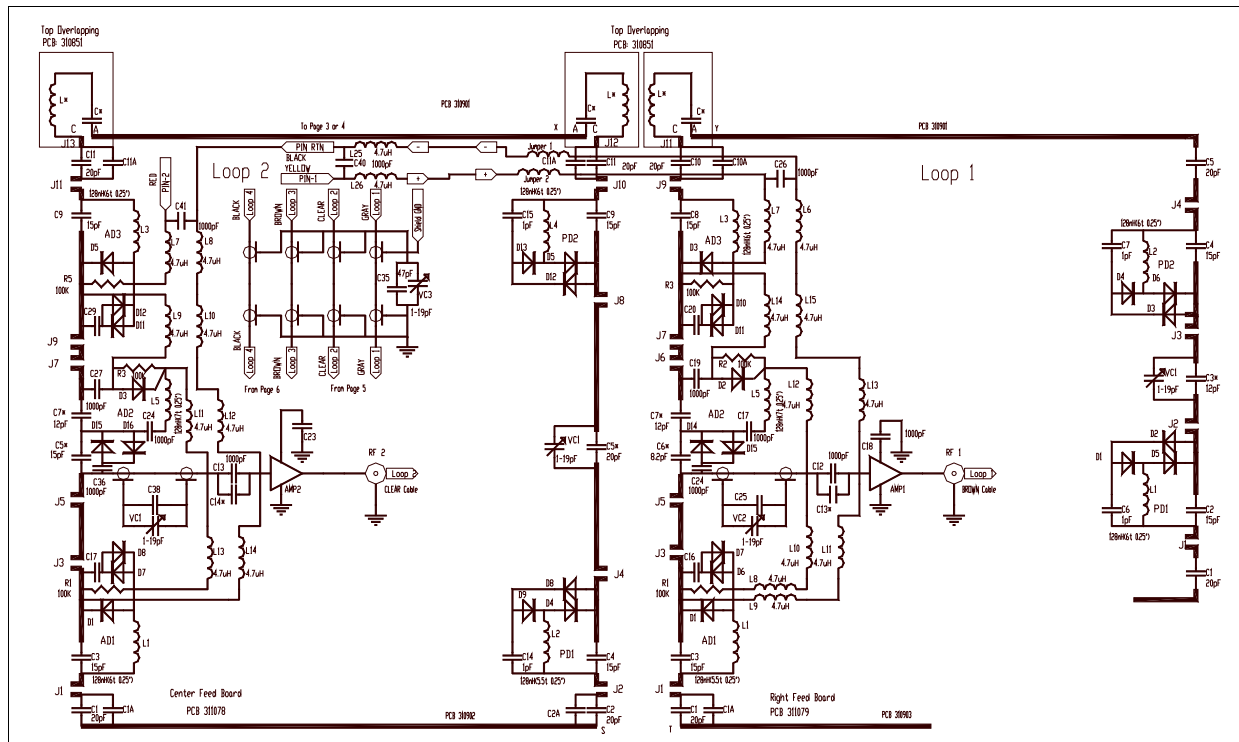
Figure 16: Parts of the coil.

SECTION 7 – APPENDIX

7-1 Schematics



Schematic of Channel 1 and Channel 2



Schematic of Channel 3 and Channel 4

7-2 Coil Configuration

Parameter	Mode
coilName	8US TORSOPA
coilCode	mb156q7appvweb2is03talpsaw63fsn5
physicalCoilDescription	GE by USAI 8 ch torso array
coilType	3
multiCoil	yes
xmitAtten	0
fastTGstartRG	12
fastTGstartTA	90
cableLoss	1.163
coilLoss	0.89
reconScale	0
autoshimRcvr	-1
headDefaultFreqDir	0
mCoilSeperationDir	7
maxFOV	480
assetThresh	10000
nuclide	1H
scic_a	1.5 0.2 0.2 2.0 128 0.4 1.25
scic_s	1.5 0.2 0.2 2.0 128 0.4 1.25
scic_c	1.5 0.2 0.2 2.0 128 0.4 1.25
mcSwitchSel	255
mcBiasEnable	255
mcVoltConfig0	255
mcVoltConfig_1	255
mcOpenCktConfig0	255
mcOpenCktConfig1	0
mcOpenCktConfig2	255
mcOpenCktConfig3	0
mcShortCktConfig0	255
mcShortCktConfig1	0
mcShortCktConfig2	255
mcShortCktConfig3	0
mcChannelSelect	255
startRec	0
endRec	7
numRec	8
userInterfaceGroup	4
prescanWeight0	1.0
prescanWeight1	1.0
prescanWeight2	1.0
prescanWeight3	1.0
prescanWeight4	1.0
prescanWeight5	1.0
prescanWeight6	1.0
prescanWeight7	1.0
aps1_mod	0
aps1_mod_PStloc	0
aps1_mod_PSrloc	0
aps1_mod_fov	200
aps1_plane	1
opthickPS_mod	5
coilPercentExposed	100
optxpos	3
extremity	no
mcReconEnable	0
mcPortEnable	6
mcErrorEnable	6
startFastRec	4
numFastRec	0
endFastRec	4
linearQuad	1

REVISION HISTORY

Date	Rev	ECO	Author	Change Description
10/03	1	N/A		First Issue
08/08	2	2008180	A. Eastman	Changed document p/n. Updated Figure 1 & 2. Table 1-1: changed Op guide p/n, added Lower Pad, changed GE p/n for Coil to N/A. Sect 1-3: changed Op Guide p/n. Section 3-2-2: Revised Step 7. Section 3-4: removed steps. Updated Warning and Caution graphics. Table 6-1 & 6-2: Updated p/n's. Updated Figure 16 and Section 7.